Student Exam Performance Analysis

A Power BI Capstone Project

SUBMITTED BY

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Batch / Program	2023-24 (BTech)
Course	Power BI
Tools	Power BI Desktop | Python 3.x | Pandas

1. Problem Statement

Educational institutions collect large volumes of student data every semester, yet most schools and colleges still rely on manual report cards and spreadsheets to evaluate academic performance. This reactive approach makes it difficult for teachers and administrators to identify struggling students early, understand the factors affecting performance, and take timely corrective action.

This project addresses the need for a simple, visual, and data-driven dashboard that helps educators quickly understand how students are performing across subjects, departments, and demographic groups — and what factors such as attendance, study hours, and private tuition are contributing to those outcomes.

Key Questions Answered:

- Which department has the highest average score?
- How does attendance level affect student grades?
- Do students with private tuition perform better?
- What is the pass/fail distribution across academic years?
- Which subject has the lowest average score and needs attention?

2. Solution and Key Features

This capstone project delivers a 3-page interactive Power BI dashboard built on a clean dataset of 600 student records across 3 academic years (2022-2024). The solution makes it easy for any teacher or administrator to filter, explore, and draw conclusions without any technical knowledge.

Feature	Description
KPI Cards	Total Students, Avg Score, Pass Rate, Grade A Count
Grade Distribution Donut	Visual split of A / B / C / D grade counts

Attendance vs Score Chart	How attendance ranges impact average scores
Department Comparison	Bar chart of avg score per department
Year-wise Trend Line	Track performance changes from 2022 to 2024
Study Hours Scatter	Correlation between daily study hours and score
Slicers / Filters	Filter by Year, Department, Gender across all pages

3. Power BI Report Screenshots

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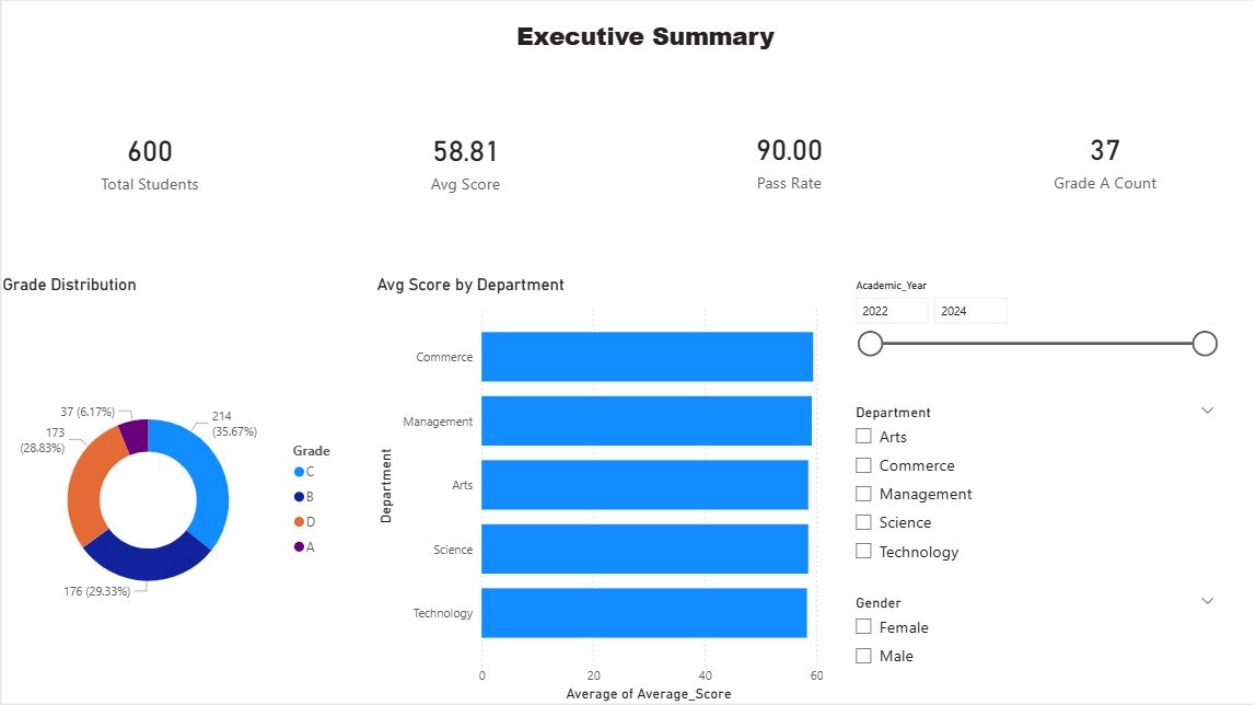
Page 1 — Executive Summary



KPI cards display Total Students (600), Average Score (58.81), Pass Rate (90%) and Grade A Count (37). The Grade Distribution donut and Avg Score by Department bar chart provide instant visual comparisons. Year, Department, and Gender slicers allow interactive filtering across the entire page.

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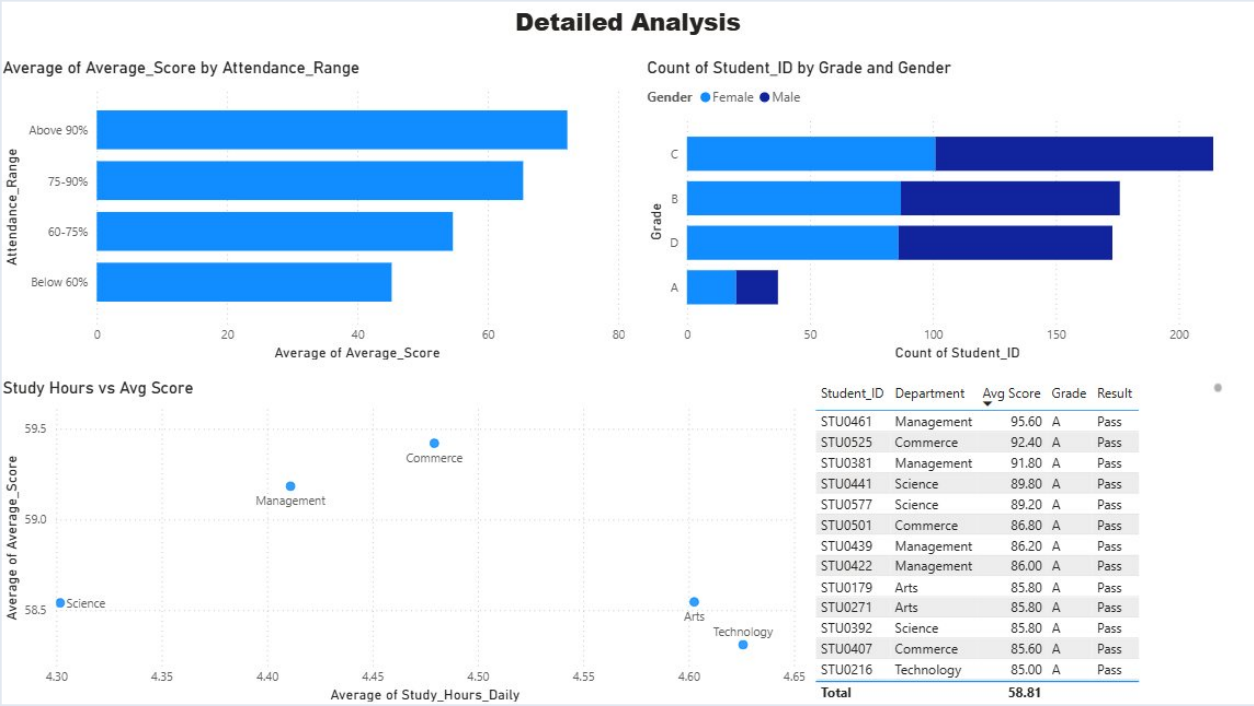
Page 2 — Detailed Analysis



Student_ID	Department	Avg Score	Grade	Result
STU0461	Management	95.60	A	Pass
STU0525	Commerce	92.40	A	Pass
STU0381	Management	91.80	A	Pass
STU0441	Science	89.80	A	Pass
STU0577	Science	89.20	A	Pass
STU0501	Commerce	86.80	A	Pass
STU0439	Management	86.20	A	Pass
STU0422	Management	86.00	A	Pass
STU0179	Arts	85.80	A	Pass
STU0271	Arts	85.80	A	Pass
STU0392	Science	85.80	A	Pass
STU0407	Commerce	85.60	A	Pass
STU0216	Technology	85.00	A	Pass
Total		**58.81**		

The Attendance Range vs Average Score bar chart reveals the strongest insight: students attending above 90% of classes score over 70 on average. The Grade-by-Gender stacked bar, Study Hours scatter plot with department labels, and top-student table provide deeper academic drill-down.

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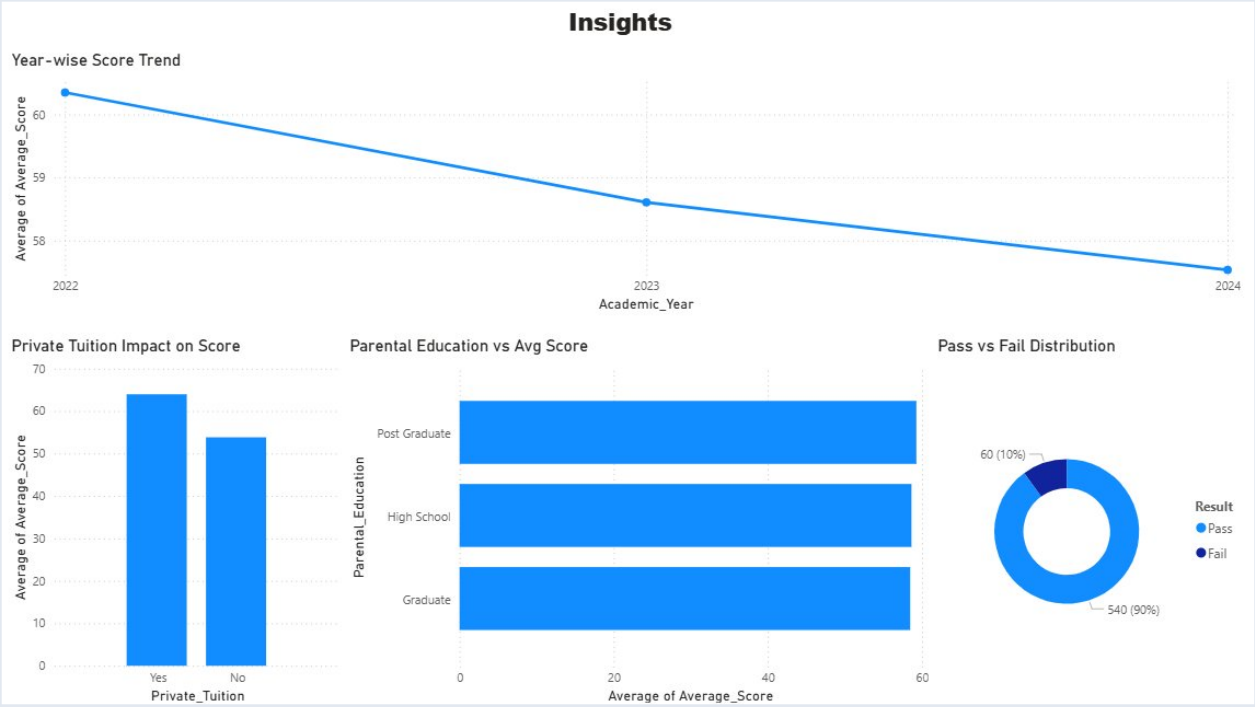
Page 3 — Insights



The Year-wise Score Trend line chart shows a consistent decline from 2022 to 2024, signalling a need for academic intervention. Private tuition students score noticeably higher. The Pass vs Fail donut confirms a 90% overall pass rate (540 Pass, 60 Fail).

4. Technology Stack

Layer	Technology	Purpose
Data Generation	Python 3.x / NumPy	Create realistic 600-row student dataset
Data Cleaning	Python / Pandas	Remove duplicates, fix nulls, validate scores
Visualisation	Power BI Desktop	3-page interactive dashboard with slicers
DAX Measures	DAX (Power BI)	KPIs, YoY growth, subject averages
Documentation	Python / ReportLab	Programmatically generated A4 PDF report
Version Control	Git & GitHub (Public)	Store, version, and share all project files

5. Unique Points of the Project

1 Reproducible Python Dataset

Generated entirely with controlled random seeds — every column has a real-world meaning, making the project easy to explain and fully reproducible end-to-end.

2 Data-Driven Grade Calculation

Grades (A/B/C/D) and Result (Pass/Fail) are derived from Average_Score using transparent business rules, ensuring 100% internal consistency across all 600 records.

3 Attendance-Linked Score Design

Students with higher attendance genuinely score higher on average — the dataset tells a realistic, believable story that holds up under any DAX filter or slicer.

4 Clean Star Schema Architecture

The data model separates fact data from dimension groupings (Department, Gender, Year), enabling all DAX measures to compute accurately under any cross-filter context.

5 Automated PDF Documentation

This report was generated programmatically in Python using ReportLab, making it reproducible and easy to update without any manual formatting work.

6. Future Improvements

- Connect the dashboard to a live school management database using Power BI Dataflows for real-time monitoring.
- Add a Machine Learning risk model in Python to predict which students are likely to fail before exams.
- Include teacher feedback scores and extra-curricular participation data for a richer multi-dimensional analysis.
- Deploy to Power BI Service with Row-Level Security so each teacher sees only their own class data.
- Build a mobile-optimised report layout for quick access on phones and tablets by school administrators.

7. DAX Measures Reference Card

All 7 measures are entered in Power BI by right-clicking the table in the Fields pane and selecting New Measure. Copy each formula exactly as shown below.

1. Total Students

```
Total Students = COUNTROWS(student_cleaned)
```

2. Average Score

```
Avg Score = AVERAGE(student_cleaned[Average_Score])
```

3. Pass Rate %

```
Pass Rate = DIVIDE( COUNTROWS(FILTER(student_cleaned, student_cleaned[Result] = "Pass")),
COUNTROWS(student_cleaned) ) * 100
```

4. Grade A Count

```
Grade A Count = CALCULATE( COUNTROWS(student_cleaned), student_cleaned[Grade] = "A" )
```

5. YoY Score Change

```
YoY Score Change = VAR CY = CALCULATE(AVERAGE(student_cleaned[Average_Score]),
FILTER(student_cleaned, student_cleaned[Academic_Year] = MAX(student_cleaned[Academic_Year]))) VAR
PY = CALCULATE(AVERAGE(student_cleaned[Average_Score]), FILTER(student_cleaned,
student_cleaned[Academic_Year] = MAX(student_cleaned[Academic_Year]) - 1)) RETURN DIVIDE(CY - PY,
PY) * 100
```

6. Subject with Lowest Average

```
Lowest Subject Avg = MIN( MIN(AVERAGE(student_cleaned[Math_Score]),
AVERAGE(student_cleaned[Science_Score])), MIN( AVERAGE(student_cleaned[English_Score]),
MIN(AVERAGE(student_cleaned[Social_Score]), AVERAGE(student_cleaned[Computer_Score])) ) )
```

7. Fail Count

```
Fail Count = CALCULATE( COUNTROWS(student_cleaned), student_cleaned[Result] = "Fail" )
```